

# Ayesha Mazhar

Computer System Engineer

---

+92 325 7068817 | ayeshamazhar332@gmail.com | linkedin.com/in/ayeshamazhar332 | Karachi, Pakistan |  
github.com/AyeshaMazhar-AM

## PROFESSIONAL SUMMARY

Computer System Engineer with hands-on internship experience building data pipelines, validating large datasets, and developing ML/DL and Generative AI applications. Reduced manual data-processing time by 70% through Python automation at the State Bank of Pakistan, and built a working Retrieval-Augmented Generation (RAG) system for document Q&A using LangChain and LLM APIs. Comfortable across the data pipeline: SQL and Python for extraction and cleaning, Scikit-learn/TensorFlow for modeling, and Power BI/Tableau concepts for reporting.

## SKILLS

- Programming & Analysis: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL
- Machine Learning: Scikit-learn, Regression & Classification, Cross-Validation, Model Evaluation (Accuracy, Precision, Recall, F1)
- Deep Learning & NLP: Neural Networks (MLP, Backpropagation), CNNs, RNNs, LSTM, TensorFlow, PyTorch, Keras, text preprocessing, tokenization, POS tagging, NLTK
- Generative AI / LLMs: LLM & RAG pipelines, LangChain, OpenAI API, Azure OpenAI, Hugging Face Transformers, Vector Databases, Prompt Engineering

## WORK EXPERIENCE

### Data Science Intern

July 2026 - Present

#### 10Pearls | Karachi, Pakistan

- Currently developing an end-to-end, serverless ML pipeline to forecast 3-day Air Quality Index (AQI) for a target city, covering automated data ingestion, feature engineering, model training, and real-time serving.
- Engineering automated feature pipelines that fetch weather and pollutant data from AQICN/OpenWeather APIs, deriving time-based and pollutant-trend features (e.g., AQI change rate) and storing them in a feature store (Hopsworks).
- Training and evaluating forecasting models (Random Forest, Ridge Regression, TensorFlow) using RMSE, MAE, and  $R^2$  metrics, with SHAP-based feature importance for model interpretability.
- Setting up hourly feature-pipeline runs and daily model retraining via Apache Airflow/GitHub Actions, and an interactive Streamlit/Flask dashboard to surface real-time predictions and hazardous AQI alerts.

### Data Science Intern

June 2026 - August 2026

#### State Bank of Pakistan | Karachi, Pakistan

- Built a modular Python 3.12+ pipeline covering financial report discovery, downloading, extraction, validation, and Excel export, automating a largely manual process for 320+ PSX company reports and reducing processing time by approximately 70%.
- Developed extraction workflows to handle Balance Sheet, Profit & Loss, and Cash Flow Statements across highly variable PDF layouts, structures, and naming conventions used by different companies — supporting both searchable-text and scanned PDFs.
- Implemented note matching, table detection, financial line-item extraction, and deterministic FSA (Financial Statement Analysis) mapping logic, along with rule-based validation to catch formatting and parsing errors before downstream analysis, improving extracted data accuracy to 85%+.
- Automated generation of structured, styled Excel workbooks from extracted data to support further financial analysis.

### Python Programming Intern

Sept 2025 - Oct 2025

#### NextGen Learners | Karachi, Pakistan

- Developed and optimized 15+ Python scripts for data analysis, preprocessing, and visualization using NumPy, Pandas, Matplotlib, and Seaborn, improving workflow efficiency.
- Operated 3 classification models (Logistic Regression, Decision Tree, Random Forest) using Scikit-learn on a 10,000+ row dataset, achieving 84% accuracy and 0.81 F1 score with the best-performing model, as measured by cross-validation.

EDUCATION

|                                                                                                                                                                                                                                                        |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <ul style="list-style-type: none"><li>• <b>B.E Computer System - CGPA 3.94 / 4.0</b><br/>Dawood University of Engineering and Technology</li><li>• <b>Intermediate (Pre-Engineering) - Grade A1</b><br/>Khursheed Govt. Girls Degree College</li></ul> | 2022 - 2026<br><br>2022 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|

PROJECTS

- Document Q&A Assistant (RAG)** - Python, LangChain, OpenAI API, Hugging Face
- Developed a Retrieval-Augmented Generation (RAG) application that answers natural-language questions over a 10+ document PDF collection, grounding responses in source content instead of relying on model memory alone.
  - Designed the ingestion and retrieval pipeline, chunking text into ~500-token segments and retrieving top-k relevant chunks via LangChain to inject as context into LLM prompts.
  - Evaluated retrieval quality across test queries, iterating on chunk size and retrieval parameters, and confirmed reduced hallucinated answers versus direct LLM queries in manual side-by-side testing.
- Customer Churn Prediction Model** - Python, Scikit-learn
- Cleaned and preprocessed a 7,000-row customer dataset (missing values, outliers, encoding) and performed EDA to identify key churn drivers, including contract type and tenure.
  - Trained and compared Logistic Regression and Random Forest classifiers via 5-fold cross-validation, achieving 87% accuracy and 0.83 F1-score on the best-performing model.

COURSES and CERTIFICATIONS

Certified Data Scientist - PITP | IBM Data Science Professional Certificate| AI: Machine Learning & Deep Learning - NAVTTC | Generative AI - Pakistan Engineering Council | Data Structures & Algorithms - Udemy | SQL for Data Science - Coursera